Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions

by

Minghan Li

A report submitted for
Science Research Project (SCIE30001)

Supervisor: A/Prof Jianzhong Qi

in the
School of Computing and Information Systems
Faculty of Engineering and Information Technology
THE UNIVERSITY OF MELBOURNE

November 2026

THE UNIVERSITY OF MELBOURNE

Abstract

School of Computing and Information Systems
Faculty of Engineering and Information Technology

Science Research Project (SCIE30001)

by Minghan Li

Machine learning is often said to herd: researchers chase what is hot, and review favours it. But a field can converge without copying. Researchers may instead stay close to a record: their own earlier work, or a topic they already know. We separate these two routes to conformity, imitation and continuity, and test the herding account where they predict different things. The data are 43,480 ICLR submissions (2017–2026), rejected and withdrawn papers included, with hypotheses registered before outcomes were read and re-tested on 11,099 held-out papers. The field did converge on foundation models, but convergence fits both routes, and how many topics look emergent depends on hindsight: 8 to 29, depending on the years a topic map is fitted on. Where the routes part, no herding is detectable. Researchers new to a topic propose methods more often than incumbents do, the reverse of our registered prediction. At review, crowded topics show neither a detectable premium nor a penalty, while papers far from their authors' earlier work are accepted less often (post hoc, confirmed on held-out papers). The conformity we can detect at ICLR points to a record, not to the crowd.

Declaration of Authorship

I, Minghan Li, declare that this report titled, 'Herded or Anchored? Testing Conformity in Machine Learning on a Decade of ICLR Submissions' and the work presented in it are my own. I confirm that:

- The report comprises only my original work towards Science Research Project (SCIE30001) except where indicated in the preface;
- due acknowledgement has been made in the text to all other material used; and
- the report is within the word limit set for it, exclusive of tables, maps, bibliographies and appendices.

Signed: _____

Date: _____

Preface

This report is the author's own work for Science Research Project (SCIE30001), carried out under the supervision of A/Prof Jianzhong Qi. It has not been submitted for another qualification and is unpublished material not submitted for publication.

Acknowledgements

I thank my supervisor, A/Prof Jianzhong Qi, for guidance and feedback throughout this project.

Contents

Abstract i

Declaration of Authorship ii

Preface iii

Acknowledgements iv

List of Figures vii

List of Tables viii

Abbreviations ix

Symbols x

1 Introduction 1

2 Two Routes to Conformity 4
2.1 Imitation . . . 4
2.2 Continuity . . . 4
2.3 Rival explanations and the timing of predictions . . . 5

3 Data and Design 6
3.1 Corpus . . . 6
3.2 Instruments . . . 6
3.3 Entry profiles . . . 7
3.4 Two distances . . . 7
3.5 Prospective measurement . . . 7
3.6 Protocol . . . 7

4 The Field: What Is Fashion and What Lasts 9
4.1 Waves . . . 9
4.2 Emergence depends on hindsight . . . 9
4.3 What lasts: the shift to foundation models . . . 10

5 Entry: Who Writes What **11**
5.1 Newcomers propose methods 11
5.2 Entering a topic, not entering ICLR 12
5.3 Farther from earlier work 13
5.4 Later uptake of foundation models 14
5.5 Open questions 14

6 The Gate: Crowd or Record? **15**
6.1 Does acceptance go with following the crowd? 15
6.2 What does acceptance go with? 17
6.3 Interpretation 18

Bibliography **19**

List of Figures

1.1 Research waves at ICLR, 2018–2026 2

4.1 Emerging topics by fitting window, and the shift to foundation models . . 10

5.1 Entry contrasts within topic and year 13

6.1 Gate tests by reference point 18

List of Tables

2.1 What each route predicts at the field, at entry and at the gate 5

5.1 Entry contrasts between newcomer and incumbent papers 12

6.1 Acceptance contrasts at the gate 16

Abbreviations

CI	Confidence interval (95% unless stated)
FM	Foundation model
ICLR	International Conference on Learning Representations
LLM	Large language model
MDE80	Minimum detectable effect at 80% power
ML	Machine learning
NLP	Natural language processing
pp	Percentage points

Symbols

$F_{\max}$	closeness to the team's earlier ICLR work	cosine
b	slope of later topic edges on earlier ones	–
k	number of topics in a topic map	–
κ	agreement between two labelling passes	–

Chapter 1

Introduction

Machine learning (ML) is often said to herd. Between 2017 and 2026, yearly submissions to the International Conference on Learning Representations (ICLR) in our discovery corpus grew 40-fold, from 384 to 15,411. The share building on a foundation model (FM) rose from 0.3% to 54.4%, and the share proposing a new architecture fell from 24.5% to 7.5%. Keywords rose and fell (Figure 1.1): generative adversarial networks fell from 8.1% of submissions in 2018, and large language models reached 16.7% in 2026. The wave is also one for natural language processing (NLP), whose researchers increasingly publish at ML venues [1]. Whether science has fashions is an old question [2]. Critics of ML scholarship point to incentives that favour short-term success [3] and to convergence on what suits the field's hardware, benchmarks and foundation models [4–6]; cascade and fashion theories explain waves by people taking their cue from others [7–9].

Is the switch to FMs not itself herding? Not necessarily: convergence on a shared tool is predicted by lock-in, by the efficient choice of a tool that works and by a bandwagon alike, so the rising curve cannot tell them apart. The herding question is about behaviour: what researchers write when they enter a topic, and what peer review accepts. Here two routes to conformity diverge. *Imitation* takes its reference point from what others are doing now: what is popular, rising or recently accepted. *Continuity* takes it from a record: the team's own earlier work, or a topic that a team member already knows. Imitation makes teams alike; continuity keeps each team near its own past. We call work that conforms by continuity *anchored*.[1]

[1] 'Anchored' names a position relative to a record, not a mechanism. It is not the anchoring-and-adjustment heuristic [10], the sense in which writing on herding in science uses the word [11].

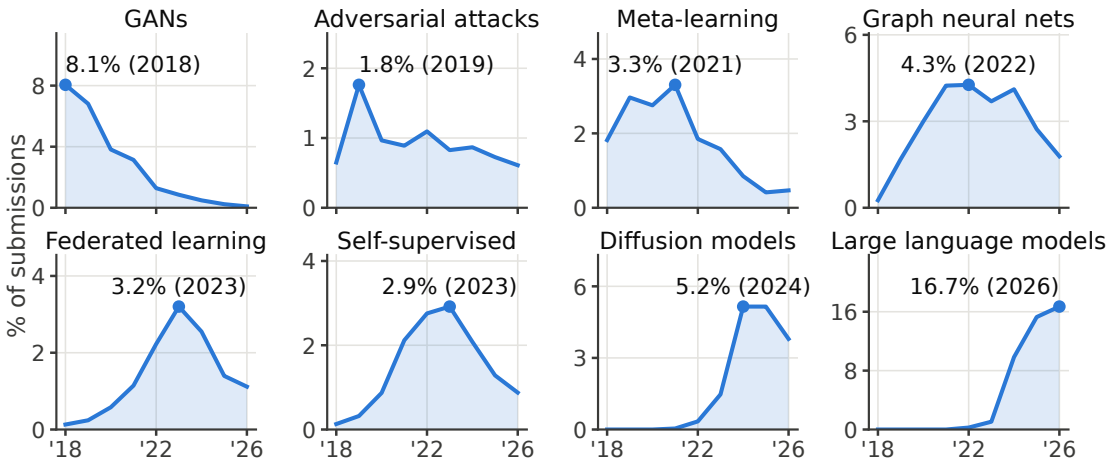


FIGURE 1.1: Research waves at ICLR: share of discovery submissions listing each author keyword, 2018–2026; dots mark the peak. The eight keywords were chosen for display after their curves were known. Across the 68 most frequent keywords, 17 peaked in 2019–2024 with shares of at most half the peak both before it and in 2026. Uses no acceptance data.

Read as herding, the waves make three claims: they are fashions; newcomers to a topic carry them by following others; and review favours following. We test each claim where it applies: in the field as a whole, at entry to a topic, and at the acceptance decision, which we call the gate. The data are 43,480 ICLR submissions from 2017–2026, rejected and withdrawn papers included. Hypotheses were registered before any full-corpus label existed, outcomes stayed sealed until one frozen run, and the main tests were run again on 11,099 held-out papers: a pre-registered out-of-sample replication, not a certified untouched holdout. Each finding carries one of three labels: REPLICATED (from a registered test, supported again on held-out papers), HELD-OUT SUPPORTED (found post hoc, then supported by a test declared before the held-out data were read) or EXPLORATORY (discovery only, or descriptive). Tests without a verdict are called inconclusive or uninformative (Chapter 3).

In the field as a whole, keyword waves are real, but how many topics look emergent depends on hindsight, and the one durable shift, the move to foundation models, fits every route. At entry, researchers new to a topic propose methods more often than incumbents, while in the FM era benchmark and evaluation work stays with incumbents. This is the opposite of the follow-on entry that herding implies, and of our own registered prediction. At the gate, no premium for crowded topics is detectable; titles that open with a coined name carry a small premium that the held-out test was too small to confirm; and most of what replicated is tied to the team's own earlier work.

Prior work relates ICLR acceptance to prestige, process and research area [12–16] and describes which papers start topics, who enters LLM research and how far researchers move from their own past work [17–19]. To our knowledge, among the works we checked, none tests the herding account at entry and at the gate within one submission population, with registered hypotheses re-tested on held-out papers, as Ibrahim et al. [13] did for a prestige gap. We contribute:

- **Entry.** A registered test of herding where imitation predicts it most sharply. Researchers entering a topic propose methods more often (+3.7 pp; REPLICATED). The surplus holds for researchers with ICLR work in other topics and predates the FM wave, while, in the FM era, benchmark and evaluation work sits with topic incumbents and larger teams (EXPLORATORY).
- **Gate.** Across every decided submission under double-blind review, rejected ones included, acceptance goes with closeness to the team's own earlier work: papers far from it are accepted 5.4 pp less often (6.1 pp on held-out papers; HELD-OUT SUPPORTED). No association with following the crowd is detectable. Hill et al. [20] relate moving away from one's own research to publication success; we add the accept/reject decision itself.
- **Method.** On the same papers, the number of emerging topics ranges from 8 to 29 depending on the years the topic map is fitted on (EXPLORATORY). Emergence is hard to judge before the fact [21, 22], and a prospective signal has been tested [23]; an early-window refit is a cheap check on trend claims.

Chapter 2

Two Routes to Conformity

2.1 Imitation

When outcomes are uncertain, organisations mimic one another [24], individuals follow their predecessors and discard their own information [7, 8], and herding protects a reputation when evaluators judge people against their peers [25]; bandwagons can spread even when adopting brings no gain [26]. In ML, benchmark use concentrates on a small set of datasets, increasingly so [27]. Entrants to a topic face the most uncertainty, so imitation predicts that they flock to hot topics, adopt rising tools early, and write follow-on work that benchmarks or evaluates what others proposed. At the gate, if review favours following, papers in crowded topics, in the popular form or riding a wave are accepted more often.

2.2 Continuity

Search stays near existing competence [28, 29], and taking up new knowledge depends on related prior knowledge [30]. At entry, continuity predicts that incumbents consolidate a topic's record and that new tools spread through teams that already have a record; accounts of outsiders in science instead expect entrants to bring work from elsewhere [31, 32]. At the gate, acceptance should track how close a paper stays to its team's record. Weisberg and Muldoon [33] separate 'followers' of others' approaches from agents who ignore them; we borrow the distinction, not their welfare conclusions [34].

Route (reference point)	Field	Entry	Gate
Imitation (others, now)	Convergence on shared tools	Entrants flock to hot topics, adopt rising tools early and write follow-on benchmark and evaluation work	A premium for crowded topics, the popular form and riding a wave
Continuity (a record)	Convergence on shared tools	Entrants bring other work; incumbents consolidate; new tools spread through teams with a record	Acceptance tracks closeness to the team's own record

TABLE 2.1: What each route predicts at the three sites. Lock-in and efficient choice also predict convergence on shared tools, so only entry and the gate can tell the routes apart.

2.3 Rival explanations and the timing of predictions

Visible status, cumulative advantage and expertise [13, 35, 36] also predict that acceptance goes with a team's record. Network diffusion, in which less connected actors adopt later [37], also predicts that teams new to ICLR take up new tools later, so that contrast cannot separate the routes. Imitation's entry and gate predictions (Table 2.1) are the herding hypotheses we registered before reading any outcome; continuity's were written after the tests had run. The distinction organises the tests; it did not generate them. To our knowledge, among the works we checked, the two routes have not been tested against each other at entry or at the gate.

Chapter 3

Data and Design

3.1 Corpus

The corpus comes from our own dated acquisitions of official OpenReview records: 43,480 discovery submissions with a usable abstract and 11,099 held-out papers. Acceptance tests use the 32,190 decided discovery papers from 2018–2026, 40.1% of them accepted, and 8,185 decided held-out papers from 2018–2026; withdrawn papers are left out.

3.2 Instruments

A *topic* is one of 100 spherical k-means clusters of SPECTER2 embeddings of title and abstract [38, 39], fitted on all years unless another map is named. An LLM with a fixed codebook gives each paper one primary contribution type: a *method paper* proposes a method, including a new architecture, and a *measurement paper* contributes a dataset, benchmark or evaluation; an *FM paper* uses or adapts a large pre-trained model. A blind second pass agreed at κ 0.93, and the author hand-checked a small random sample. This is consistency between LLM labellings, not human-level validity; zero-shot LLM labels reach only fair agreement with human coders on many tasks [40].

3.3 Entry profiles

Entry status is relative to a topic and counts ICLR submissions only. In a *newcomer paper*, no author has an earlier ICLR submission, with any outcome, in the paper's topic (a topic-relative definition, as in Biliotti et al. [41]); otherwise it is an *incumbent paper*, and an author with such a submission is a *topic insider*. A newcomer paper comes from a *first-time team* if no author has any earlier ICLR submission, and from a *returning topic newcomer* if some author has one in another topic.

3.4 Two distances

Both distances use one centred SPECTER2 space, with different reference points. *Closeness to the team's own record*, $F_{\max}$, is the highest cosine between the paper and the mean of an author's earlier ICLR submissions since 2021, taken over the paper's *experienced authors*, those with such submissions, following Bao and Li [19]. *Distance from earlier ICLR work* is one minus the mean cosine to the ten nearest submissions of earlier years, as a percentile within topic and year.

3.5 Prospective measurement

Predictors use only the past: *prior FM adoption* is the FM share of the topic's previous-year papers, *crowding* is the log of one plus the topic's earlier papers, and a *hot topic* is in the top fifth of prior growth. Code links leak the outcome into 17.6% of accepted but 0.5% of rejected 2023–2026 abstracts [42]; in discovery, the distance-based gate estimates hold without them (no link-removed text exists for held-out papers).

3.6 Protocol

Hypotheses, models and a support rule were registered before any full-corpus label existed. A hypothesis is supported if it has the predicted sign, a Holm-adjusted $p < 0.05$ in its family [43], and that sign in at least 75% of counted variants and in both hindsight

variants (a year-balanced map; the 2017–2020 map applied to 2021–2026 papers). Outcomes could be read only by committed code in one logged, gated run [44, 45]. Post hoc analyses were declared before they were run, and their one-sided held-out tests before any held-out estimate for that question was computed. Models are linear probability models with topic and year fixed effects and topic-clustered errors. We disclose two weaknesses. Under a simulated null whose structure cuts across topics, the main entry tests rejected 12.6% and 10.0% of the time at a nominal 5%, which a wild cluster bootstrap [46] did not fix. And keyword popularity and title style had been explored against acceptance on the discovery papers before registration: the discovery estimate for titles is not a test, and the keyword-timing tests probe a dimension already seen. A test that is not supported and whose estimate lies below its minimum detectable effect at 80% power (MDE80) is *uninformative*: it shows neither an association nor its absence. Supported held-out tests confirm a direction, not a size.

Chapter 4

The Field: What Is Fashion and What Lasts

Keyword waves are real, but how many topics look emergent depends on hindsight, and the one durable shift, the move to foundation models, fits every route.

4.1 Waves

Author keywords need no topic map, and at their level the waves are real: of the 68 most frequent keywords, 17 rose and fell, peaking in 2019–2024 with shares of at most half the peak both before it and in 2026 (Figure 1.1; EXPLORATORY).

4.2 Emergence depends on hindsight

Histories of the field usually fit one topic model to the whole archive [47–49], and dynamic topic models and BERTopic fit all years at once [50, 51]. We call a topic *emerging* if its 2024–2026 share is at least three times its 2017–2020 share. On the same papers and pipeline, a map fitted on 2017–2020 finds 8 emerging topics and a map fitted on all years finds 29 (Figure 4.1a; EXPLORATORY). Two reasons combine, and we cannot separate them. A map fitted on all years leaks the future into the past, like temporal leakage in prediction [52] and hindsight in judgement [53]: with 69% of papers from 2024–2026, it is in effect a present-day map. And the early map cannot represent topics born after

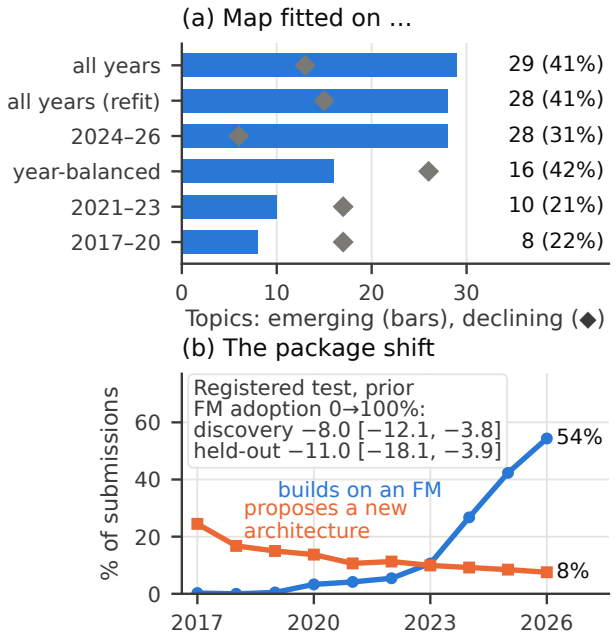


FIGURE 4.1: (a) Emerging (bars) and declining (grey diamonds) topics on $k = 100$ maps fitted on different years, same 43,480 papers; right-hand labels: emerging count (share of 2024–2026 papers in emerging topics, 22.3% on the 2017–2020 map against 40.9% on the all-years map). All years (refit): the all-years map refitted with another seed. Topic size 'predicts' two-year growth only on the all-years map (R^2 0.28 against −0.03; no paired test). (b) Share of submissions building on an FM (blue circles) and proposing a new architecture (orange squares); LLM labels. Inset: the registered test of the shift to FMs, REPLICATED (held-out confirms direction, not size). Uses no acceptance data.

2020. The two counts therefore bracket the number of emerging topics rather than fix it.

4.3 What lasts: the shift to foundation models

The FM share rose every year from 2018 (Figure 4.1b). Where more of a topic's previous-year papers built on FMs, fewer of its papers proposed new architectures: −7.98 pp [−12.14, −3.82] for prior adoption from 0 to 100% in discovery and −11.0 [−18.1, −3.9] on held-out papers (REPLICATED; the held-out test confirms the direction, not the size). No detectable shift towards empirical or evaluation papers came with it (−0.63 [−3.64, +2.38]; not supported, inconclusive), and the association may also run the other way. The pattern matches accounts of NLP's paradigm shifts [54–56]. But lock-in [4], efficient choice [57] and a bandwagon [58] all predict such convergence, so the field level cannot tell imitation from continuity; only entry and the gate can.

Chapter 5

Entry: Who Writes What

Entry shows a division of labour: researchers entering a topic bring methods, while, in the FM era, benchmark and evaluation work sits with larger teams and topic incumbents. This is the opposite of the follow-on entry that the herding account implies.

5.1 Newcomers propose methods

We registered that newcomer papers, whose authors face the most uncertainty, would more often be follow-on work: benchmarks, datasets and evaluations of what others had proposed. Newcomer papers are 72.9% of the 42,325 papers from 2019–2026 with an author identifier. They are more often method papers, +3.71 pp [+2.61, +4.80] in discovery and +3.60 [+1.48, +5.71] held-out, and less often measurement papers, −1.70 [−2.46, −0.93] and −2.10 [−3.35, −0.85] (Table 5.1, Figure 5.1; REPLICATED). Our registered prediction thus failed in the opposite direction, and the reversed hypotheses, registered before any held-out data existed, held on held-out papers. All 10 counted variants agree, and the t-statistics (6.7 and −4.4) are large against the inflated test size. Much of the measurement gap is accounted for by team size: with log team size it shrinks to −0.54 [−1.11, +0.04] in discovery and −0.98 held-out, while the method gap does not move (+3.92 and +3.64).

Contrast	Discovery [95% CI]	Held-out [95% CI]	Label
Method paper	+3.71 [+2.61, +4.80]	+3.60 [+1.48, +5.71]	REPLICATED
Measurement paper	−1.70 [−2.46, −0.93]	−2.10 [−3.35, −0.85]	REPLICATED
Measurement paper, with log team size[a]	−0.54 [−1.11, +0.04]	−0.98	registered variant
Method: returning topic newcomer − incumbent[b]	+3.43 [+2.31, +4.55]	–	EXPLORATORY
Method: first-time team − returning topic newcomer[b]	+0.92 [−0.44, +2.29]	–	EXPLORATORY
Distance from earlier ICLR work, own work excluded (percentile points)	+4.33 [+3.44, +5.22]	–	EXPLORATORY
Flocking: incumbent share, hot − other topics	−0.73 [−2.68, +1.22]	not registered	inconclusive

TABLE 5.1: Entry: newcomer minus incumbent papers unless the row names other profiles; pp within topic and year. Held-out tests are one-sided and confirm direction, not size. [a]Registered supplementary variant, outside the support rule. [b]First-time team: no author has any earlier ICLR submission; returning topic newcomer: an author has one, in another topic.

5.2 Entering a topic, not entering ICLR

Newcomer papers split into first-time teams (26.3% of papers; median team of 3) and returning topic newcomers (46.7%; median 5); incumbent papers are the other 27.0% (median 5). Returning topic newcomers write method papers more often than incumbents, +3.43 pp [+2.31, +4.55], at the same median team size and also with team size controlled (+3.62). First-time teams do so by +4.35 [+2.82, +5.89], and the two newcomer profiles do not differ detectably (+0.92 [−0.44, +2.29]; all EXPLORATORY). The method surplus thus goes mainly with entering a topic, not with being new to ICLR or with small teams, and it is present in every era, so it predates the FM wave. The measurement gap is different. For first-time teams it is accounted for by team size (−2.24 [−3.24, −1.25] against incumbents, +0.09 [−0.63, +0.80] with team size); for returning topic newcomers part of it remains (−1.47, and −0.81 [−1.46, −0.16] with team size). It appears only in 2024–2026 (first-time teams against incumbents −2.75 [−4.07, −1.44]). In the FM era, then, benchmark and evaluation work sits with topic incumbents and larger teams; team size is the only resource we measure. NLP shows a parallel at the level of research trends, where tasks and methods lead and datasets follow [56].

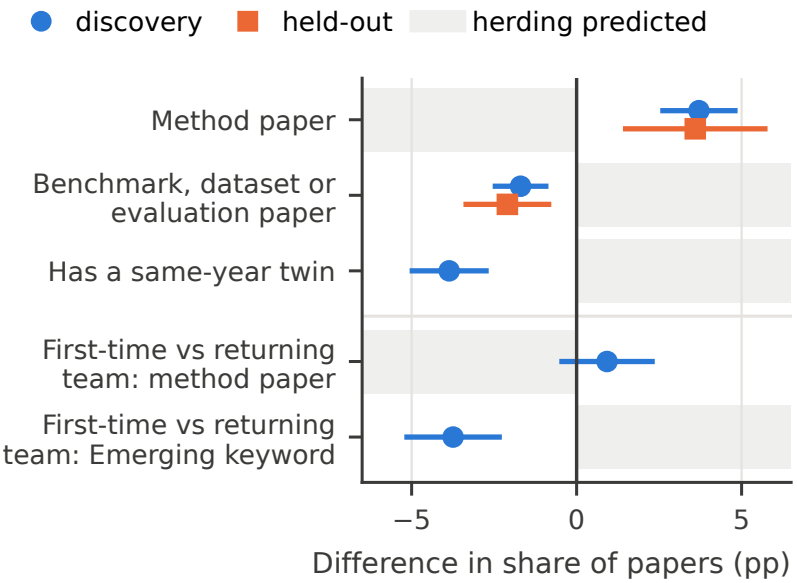


FIGURE 5.1: Entry contrasts within topic and year (pp) with 95% intervals: discovery (blue circles) and held-out (orange squares). Rows 1–2, newcomer minus incumbent papers, are registered and REPLICATED (held-out confirms direction, not size). Rows 3–5 are EXPLORATORY (discovery only): row 3 compares papers by topic newcomers with an ICLR record with incumbent papers (first-time teams controlled separately; class, FM and team-size controls); row 4 compares first-time teams with returning topic newcomers, and row 5 with all teams that have an ICLR record. Grey: the side the herding prediction expected. Uses no acceptance data.

5.3 Farther from earlier work

Newcomer papers also sit farther from earlier ICLR work: +5.90 percentile points [+4.95, +6.85] within topic and year, +5.24 among method papers alone, and +4.33 [+3.44, +5.22] with the authors' own earlier papers excluded from the reference set (EXPLORATORY). Papers by topic newcomers with an ICLR record are also less likely than incumbent papers to have a same-year *twin*, an independent submission at cosine ≥ 0.70, with first-time teams controlled separately (-3.86 pp [-4.98, -2.74]; EXPLORATORY). The follow-on prediction expects newcomer papers to sit closer to earlier work, not farther. Writing method papers could itself be imitation of the venue's dominant genre, since three quarters of ICLR papers are method papers, but newcomers' distance from earlier work tells against that reading. A method paper is a contribution type, and embedding distance is not originality: these measures give no sign that newcomers follow, and do not show that they are more original.

5.4 Later uptake of foundation models

In 2023–2026 topics at 5–50% prior FM adoption, first-time teams write FM papers less often than returning topic newcomers, by −4.58 pp [−6.55, −2.60] up to −14.71 [−19.82, −9.61] as adoption rises, while returning topic newcomers do not differ detectably from incumbents. First-time teams also list young, rising *Emerging* keywords less often than teams with an ICLR record: −3.74 [−5.14, −2.35] (both EXPLORATORY). Foundation models thus spread through teams with an ICLR record, as absorptive capacity predicts [30]; network diffusion predicts the same order, so this contrast does not discriminate.

5.5 Open questions

Whether entrants crowd into hot topics is unsettled: the registered flocking test, the incumbent share in hot against other topics, is inconclusive (−0.73 [−2.68, +1.22]). At entry, no detectable difference fits the follow-on prediction.

Chapter 6

The Gate: Crowd or Record?

At the gate we detect no premium for following the crowd. Apart from topic-level differences in acceptance, which persist only in part, what replicated is tied to the team's own earlier work. Continuity predicts this, but so do status and expertise.

6.1 Does acceptance go with following the crowd?

The registered crowding test predicted a penalty, and it is uninformative: −1.68 pp per log unit (natural log of one plus the topic's earlier papers) [−3.68, +0.32] in discovery and −1.51 [−5.20, +2.19] held-out, against an MDE80 of 4.8 (Table 6.1, Figure 6.1). Neither a penalty nor a premium is detectable, and held-out premiums above +2.19 pp per log unit lie outside the interval.

The plainest imitation-like spread in the corpus is in form. The share of titles that open with a short coined name and a colon rose from 10.9% in 2017 to 44.6% in 2026. Within topic and year, with class, FM and team-size controls, such branded titles are accepted +1.74 pp [+0.37, +3.11] more often in discovery, which is not a test, since title style had been explored before registration. The declared held-out test gives +1.76 [−0.82, +4.34] (one-sided p 0.089, MDE80 3.2): not supported, and uninformative, so the premium stays EXPLORATORY. Raw within-year gaps are larger (+3.3 in discovery, +4.0 held-out); about half goes with topic, type, FM use and team size. Titles can also be revised after the decision, which could create such a premium.

Acceptance and ...	Discovery [95% CI]	Held-out [95% CI]	Label
Others: crowded topics, popular title forms, foundation models			
Crowding, per log unit[a]	−1.68 [−3.68, +0.32]	−1.51 [−5.20, +2.19]	uninformative
Branded title[b]	+1.74 [+0.37, +3.11]	+1.76 [−0.82, +4.34]	EXPLORATORY
FM premium, per logit unit of prior adoption	+0.19 [−0.96, +1.34]	–	EXPLORATORY
Topic edge 2018–22 → 2024–26, slope[c]	0.38 [0.18, 0.58]	0.60 [0.32, 0.88]	HELD-OUT SUPPORTED
Emerging − Mainstream keyword[b,c]	+2.67 [+0.76, +4.58]	+3.25 [−0.18, +6.69]	EXPLORATORY
The team's own record			
Far jump: $F_{\max}$ p10 − median	−5.44 [−7.10, −3.78]	−6.09 [−9.32, −2.86]	HELD-OUT SUPPORTED
Same author, +0.1 cosine closer[c]	+0.55 [+0.21, +0.90]	+1.06 [+0.23, +1.89]	HELD-OUT SUPPORTED
One topic insider vs none[c]	+3.42 [+1.18, +5.66]	+3.88 [+0.28, +7.47]	HELD-OUT SUPPORTED
The field's record			
Least-distant fifth − middle fifth	−3.27 [−4.93, −1.60]	−0.13 [−3.50, +3.25]	uninformative

TABLE 6.1: The gate: pp within topic and year (topic edges and keywords: against the same year's other papers). Rows differ in population and unit: compare replication status, not size. Held-out tests are one-sided and confirm direction, not size. [a]Registered; predicted a penalty; MDE80 4.8. [b]Explored against acceptance before registration. Titles: the discovery estimate is not a test; held-out MDE80 3.2. Keywords: the discovery estimate is a declared post hoc test on a dimension already seen. [c]Holm across the six secondary held-out tests: topic edge 0.019, same author 0.026, insider 0.052, Emerging keywords 0.063.

FM papers are accepted more often, +4.62 pp [+2.79, +6.44], with no detectable difference between late and early adopters: the premium changes by +0.19 [−0.96, +1.34] per logit unit (log-odds of the topic's prior FM adoption; EXPLORATORY), which fits rewarded imitation and efficient choice alike. A topic's *edge*, its acceptance minus that of all other papers of the same year (topics of the 2017–2020 map), is real but regresses: the 2018–2022 edge predicts the 2024–2026 edge with slope 0.38 [0.18, 0.58] among discovery papers, and the held-out papers' 2024–2026 edges regress on the discovery 2018–2022 edges with slope 0.60 [0.32, 0.88] (HELD-OUT SUPPORTED; two different comparisons). A favoured topic keeps part of its edge, not all of it, as expected when decisions carry a large arbitrary component [59, 60]. Keyword timing is weak (EXPLORATORY): on held-out papers, *Emerging* keywords go with higher acceptance than old, flat *Mainstream* ones (+3.25 [−0.18, +6.69]), and acceptance may fall after a keyword's share peaks (−5.2 [−11.1, +0.7]; −1.7 without the six keywords inspected first). Neither shows the flat premium that rewarded imitation of what is popular predicts.

6.2 What does acceptance go with?

The clearest pattern is the *far jump*: how much less often papers far from the team's earlier work are accepted. Among 20,430 decided 2023–2026 papers with an experienced author, acceptance at the 10th percentile of closeness to the team's own record minus that at the median is −5.44 pp [−7.10, −3.78] in discovery and −6.09 [−9.32, −2.86] held-out, with power 0.89 (HELD-OUT SUPPORTED; the declared primary test). Adjusted acceptance rises with closeness up to the fourth fifth. The far jump holds with a newcomer control (−5.0), although closeness correlates −0.57 with the newcomer flag. Holding the author fixed, a paper 0.1 closer in cosine to that author's earlier work goes with acceptance +0.55 pp [+0.21, +0.90] higher in discovery and +1.06 [+0.23, +1.89] held-out (HELD-OUT SUPPORTED; it survives a family-wise correction across the secondary tests). Controls for newcomer status and for co-authors' experience move it little (+0.48 and +0.50), though unmeasured differences in co-author quality could also bias it, as a simulation run before the test showed. Likewise, the same author is accepted about 1.4 pp less often [−2.74, −0.004] on a paper in a topic new to them (author fixed effects; EXPLORATORY).[1]

One topic insider on the team, against none, goes with +3.42 pp [+1.18, +5.66] in discovery and +3.88 [+0.28, +7.47] held-out, at fixed team size and number of experienced authors (HELD-OUT SUPPORTED). It does not survive the family-wise correction (Holm 0.052), and a status reading fits it as well: per head, an insider goes with only +0.58 pp more than an experienced author from another topic (p 0.16; EXPLORATORY).

The field's record is undetermined (uninformative). In the same space, the least-distant fifth of papers against the middle fifth is −3.27 [−4.93, −1.60] in discovery but −0.13 [−3.50, +3.25] held-out, with power 0.47 (uninformative), and the within-author slope for distance from the field was not supported either. Of the distance-based tests, only the own-record associations replicated.

[1] Newcomer papers as a whole are accepted 10.2 pp less often [−11.8, −8.7] within topic and year (EXPLORATORY), but that gap mostly reflects who writes them: the gap shrinks once team size and the authors' ICLR experience are controlled, and further within the same authors.

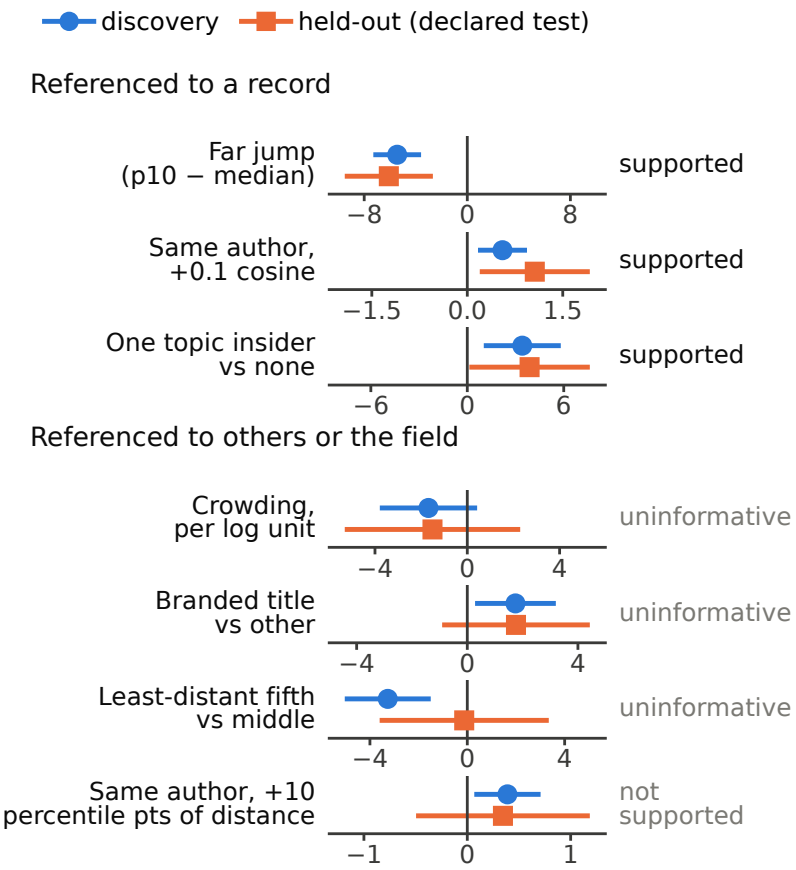


FIGURE 6.1: Gate tests by reference point: discovery estimates (blue circles) and held-out declared tests (orange squares), 95% intervals; the held-out verdict is at right (supported: HELD-OUT SUPPORTED; uninformative: not supported and below the test's MDE80; not supported: a declared test without a registered MDE80). Each row has its own scale: pp for the far jump, insider, branded title and least-distant fifth; pp per log unit for crowding; per 0.1 cosine and per 10 percentile points for the two same-author slopes. Compare replication status, not size. Crowding was registered; title style had been explored before registration; the other rows were found post hoc in discovery. Each held-out test was declared before any held-out estimate for that question was computed.

6.3 Interpretation

These associations are referenced to a record, as continuity predicts, but expertise, evaluator conservatism and visible status predict them as well [35, 36]. Double-blind review does not rule status out: where reviewers see the authors, famous authors fare better [61], and preprints can reveal who wrote a paper [62]. Unlike studies of published work [20], these results come from the accept/reject decision for every decided submission.

Bibliography

[1] David Jurgens. The future of NLP may not be at NLP conferences: Scholarly migration patterns in natural language processing. arXiv preprint arXiv:2607.02416, 2026.

[2] Diana Crane. Fashion in science: Does it exist? *Social Problems*, 16(4):433–441, 1969. doi: 10.1525/sp.1969.16.4.03a00040.

[3] Zachary C. Lipton and Jacob Steinhardt. Troubling trends in machine learning scholarship: Some ML papers suffer from flaws that could mislead the public and stymie future research. *Queue*, 17(1):45–77, 2019. doi: 10.1145/3317287.3328534.

[4] Sara Hooker. The hardware lottery. *Communications of the ACM*, 64(12):58–65, 2021.

[5] Mostafa Dehghani, Yi Tay, Alexey A. Gritsenko, Zhe Zhao, Neil Houlsby, Fernando Diaz, Donald Metzler, and Oriol Vinyals. The benchmark lottery. arXiv preprint arXiv:2107.07002, 2021.

[6] Rishi Bommasani et al. On the opportunities and risks of foundation models. arXiv preprint arXiv:2108.07258, 2021.

[7] Sushil Bikhchandani, David Hirshleifer, and Ivo Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

[8] Abhijit V. Banerjee. A simple model of herd behavior. *Quarterly Journal of Economics*, 107(3):797–817, 1992.

[9] Eric Abrahamson. Managerial fads and fashions: The diffusion and rejection of innovations. *Academy of Management Review*, 16(3):586–612, 1991.

[10] Amos Tversky and Daniel Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974. doi: 10.1126/science.185.4157.1124.

[11] Michelle Baddeley. Herding, social influences and behavioural bias in scientific research. *EMBO Reports*, 16(8):902–905, 2015. doi: 10.15252/embr.201540637.

[12] David Tran, Alex Valtchanov, Keshav Ganapathy, Raymond Feng, Eric Slud, Micah Goldblum, and Tom Goldstein. An open review of OpenReview: A critical analysis of the machine learning conference review process. arXiv preprint arXiv:2010.05137, 2020.

[13] Hazem Ibrahim, Talal Rahwan, and Yasir Zaki. Bias at the borderline: Who gets the benefit of the doubt in peer review? arXiv preprint arXiv:2607.26280, 2026.

[14] Sangkeun Jung, Goun Pyeon, Inbum Heo, and Hyungjin Ahn. What drives paper acceptance? A process-centric analysis of modern peer review. arXiv preprint arXiv:2509.25701, 2025.

[15] Gang Wang, Qi Peng, Yanfeng Zhang, and Mingyang Zhang. What have we learned from OpenReview? *World Wide Web*, 26:683–708, 2023. doi: 10.1007/s11280-022-01109-z.

[16] Binyan Xu et al. Reviewer scores are not comparable across research areas in ML peer review. arXiv preprint arXiv:2607.27209, 2026.

[17] Fan Huang. Catalyst papers in artificial intelligence research: A landscape on ICLR from 2017 to 2025. arXiv preprint arXiv:2607.05401, 2026.

[18] Rajiv Movva, Sidhika Balachandar, Kenny Peng, Gabriel Agostini, Nikhil Garg, and Emma Pierson. Topics, authors, and institutions in large language model research: Trends from 17K arXiv papers. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 1223–1243, Mexico City, Mexico, 2024. Association for Computational Linguistics. URL https://aclanthology.org/2024.naacl-long.67.

[19] Honglin Bao and Kai Li. Persistence paradox in dynamic science. arXiv preprint arXiv:2506.22729, 2025.

[20] Ryan Hill, Yian Yin, Carolyn Stein, Xizhao Wang, Dashun Wang, and Benjamin F. Jones. The pivot penalty in research. *Nature*, 642(8069):999–1006, 2025. doi: 10.1038/s41586-025-09048-1.

[21] Daniele Rotolo, Diana Hicks, and Ben R. Martin. What is an emerging technology? *Research Policy*, 44(10):1827–1843, 2015. doi: 10.1016/j.respol.2015.06.006.

[22] Xue Li, Ciro D. Esposito, Paul Groth, Jonathan Sitruk, Balazs Szatmari, and Nachoem Wijnberg. Evaluation of unsupervised static topic models' emergence detection ability. *PeerJ Computer Science*, 11:e2875, 2025. doi: 10.7717/peerj-cs.2875.

[23] Rasul Khanbayov and Hasan Kurban. Topical phase transitions in artificial intelligence research: Large-scale evidence and an early-warning signature for emerging topics. arXiv preprint arXiv:2606.12828, 2026.

[24] Paul J. DiMaggio and Walter W. Powell. The iron cage revisited: Institutional isomorphism and collective rationality in organizational fields. *American Sociological Review*, 48(2):147–160, 1983. doi: 10.2307/2095101.

[25] David S. Scharfstein and Jeremy C. Stein. Herd behavior and investment. *American Economic Review*, 80(3):465–479, 1990.

[26] Eric Abrahamson and Lori Rosenkopf. Institutional and competitive bandwagons: Using mathematical modeling as a tool to explore innovation diffusion. *Academy of Management Review*, 18(3):487–517, 1993. doi: 10.2307/258906.

[27] Bernard Koch, Emily Denton, Alex Hanna, and Jacob G. Foster. Reduced, reused and recycled: The life of a dataset in machine learning research. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021. URL https://datasets-benchmarks-proceedings.neurips.cc/paper/2021/hash/3b8a614226a953a8cd9526fca6fe9ba5-Abstract-round2.html.

[28] James G. March. Exploration and exploitation in organizational learning. *Organization Science*, 2(1):71–87, 1991. doi: 10.1287/orsc.2.1.71.

[29] Toby E. Stuart and Joel M. Podolny. Local search and the evolution of technological capabilities. *Strategic Management Journal*, 17(S1):21–38, 1996. doi: 10.1002/smj.4250171004.

[30] Wesley M. Cohen and Daniel A. Levinthal. Absorptive capacity: A new perspective on learning and innovation. *Administrative Science Quarterly*, 35(1):128–152, 1990. doi: 10.2307/2393553.

[31] An Zeng, Ying Fan, Zengru Di, Yougui Wang, and Shlomo Havlin. Fresh teams are associated with original and multidisciplinary research. *Nature Human Behaviour*, 5(10):1314–1324, 2021. doi: 10.1038/s41562-021-01084-x.

[32] Mikko Packalen and Jay Bhattacharya. Age and the trying out of new ideas. *Journal of Human Capital*, 13(2):341–373, 2019. doi: 10.1086/703160.

[33] Michael Weisberg and Ryan Muldoon. Epistemic landscapes and the division of cognitive labor. *Philosophy of Science*, 76(2):225–252, 2009. doi: 10.1086/644786.

[34] Jason McKenzie Alexander, Johannes Himmelreich, and Christopher Thompson. Epistemic landscapes, optimal search, and the division of cognitive labor. *Philosophy of Science*, 82(3):424–453, 2015. doi: 10.1086/681766.

[35] Robert K. Merton. The Matthew effect in science. *Science*, 159(3810):56–63, 1968. doi: 10.1126/science.159.3810.56.

[36] Danielle Li. Expertise versus bias in evaluation: Evidence from the NIH. *American Economic Journal: Applied Economics*, 9(2):60–92, 2017. doi: 10.1257/app.20150421.

[37] James Coleman, Elihu Katz, and Herbert Menzel. The diffusion of an innovation among physicians. *Sociometry*, 20(4):253–270, 1957.

[38] Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2270–2282, Online, 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.207.

[39] Amanpreet Singh, Mike D'Arcy, Arman Cohan, Doug Downey, and Sergey Feldman. SciRepEval: A multi-format benchmark for scientific document representations. In

Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing, pages 5548–5566, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.338.

[40] Caleb Ziems, William Held, Omar Shaikh, Jiaao Chen, Zhehao Zhang, and Diyi Yang. Can large language models transform computational social science? *Computational Linguistics*, 50(1):237–291, 2024. doi: 10.1162/coli_a_00502.

[41] Carolina Biliotti, Luca Verginer, and Massimo Riccaboni. Breaking new ground, reinforcing old gaps: Gender disparities in access to emerging research frontiers. arXiv preprint arXiv:2404.04707, 2024.

[42] Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15:1–15:21, 2012. doi: 10.1145/2382577.2382579.

[43] Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.

[44] Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2606, 2018. doi: 10.1073/pnas.1708274114.

[45] Olmo R. van den Akker, Sara Weston, Lorne Campbell, et al. Preregistration of secondary data analysis: A template and tutorial. *Meta-Psychology*, 5, 2021. doi: 10.15626/MP.2020.2625.

[46] A. Colin Cameron, Jonah B. Gelbach, and Douglas L. Miller. Bootstrap-based improvements for inference with clustered errors. *The Review of Economics and Statistics*, 90(3):414–427, 2008. doi: 10.1162/rest.90.3.414.

[47] David Hall, Daniel Jurafsky, and Christopher D. Manning. Studying the history of ideas using topic models. In *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, pages 363–371, Honolulu, Hawaii, 2008. Association for Computational Linguistics. URL https://aclanthology.org/D08-1038.

[48] Ashton Anderson, Dan Jurafsky, and Daniel A. McFarland. Towards a computational history of the ACL: 1980–2008. In *Proceedings of the ACL-2012 Special Workshop on Rediscovering 50 Years of Discoveries*, pages 13–21, Jeju Island, Korea, 2012. Association for Computational Linguistics. URL https://aclanthology.org/W12-3202/.

[49] Joel Klinger, Juan Mateos-Garcia, and Konstantinos Stathoulopoulos. A narrowing of AI research? arXiv preprint arXiv:2009.10385, 2020.

[50] David M. Blei and John D. Lafferty. Dynamic topic models. In *Proceedings of the 23rd International Conference on Machine Learning (ICML '06)*, volume 148 of *ACM International Conference Proceeding Series*, pages 113–120. ACM, 2006. doi: 10.1145/1143844.1143859.

[51] Maarten Grootendorst. BERTopic: Neural topic modeling with a class-based TF-IDF procedure. arXiv preprint arXiv:2203.05794, 2022.

[52] Sayash Kapoor and Arvind Narayanan. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804, 2023. doi: 10.1016/j.patter.2023.100804.

[53] Baruch Fischhoff. Hindsight ≠ foresight: The effect of outcome knowledge on judgment under uncertainty. *Journal of Experimental Psychology: Human Perception and Performance*, 1(3):288–299, 1975. doi: 10.1037/0096-1523.1.3.288.

[54] Sireesh Gururaja, Amanda Bertsch, Clara Na, David Gray Widder, and Emma Strubell. To build our future, we must know our past: Contextualizing paradigm shifts in natural language processing. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13310–13325, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.822. URL https://aclanthology.org/2023.emnlp-main.822.

[55] Naomi Saphra, Eve Fleisig, Kyunghyun Cho, and Adam Lopez. First tragedy, then parse: History repeats itself in the new era of large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 2310–2326, Mexico City, Mexico, 2024. Association for Computational Linguistics. doi: 10.18653/v1/2024.naacl-long.128.

[56] Aniket Pramanick, Yufang Hou, Saif M. Mohammad, and Iryna Gurevych. A diachronic analysis of paradigm shifts in NLP research: When, how, and why? In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 2312–2326, Singapore, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.142. URL https://aclanthology.org/2023.emnlp-main.142.

[57] Richard S. Sutton. The bitter lesson. http://www.incompleteideas.net/IncIdeas/BitterLesson.html, March 2019.

[58] Joan H. Fujimura. The molecular biological bandwagon in cancer research: Where social worlds meet. *Social Problems*, 35(3):261–283, 1988.

[59] Corinna Cortes and Neil D. Lawrence. Inconsistency in conference peer review: Revisiting the 2014 NeurIPS experiment. arXiv preprint arXiv:2109.09774, 2021.

[60] Alina Beygelzimer, Yann N. Dauphin, Percy Liang, and Jennifer Wortman Vaughan. Has the machine learning review process become more arbitrary as the field has grown? The NeurIPS 2021 consistency experiment. arXiv preprint arXiv:2306.03262, 2023.

[61] Andrew Tomkins, Min Zhang, and William D. Heavlin. Reviewer bias in single- versus double-blind peer review. *Proceedings of the National Academy of Sciences*, 114(48):12708–12713, 2017. doi: 10.1073/pnas.1707323114.

[62] Homanga Bharadhwaj, Dylan Turpin, Animesh Garg, and Ashton Anderson. De-anonymization of authors through arXiv submissions during double-blind review. arXiv preprint arXiv:2007.00177, 2020.